Definition 1.2 (Domain): A domain D is a non-trivial, connected, open subset of $\mathbf{R}^d$. D is said to be *bounded* if $D \subset \{\mathbf{x} \in \mathbf{R}^d : \|\mathbf{x}\|_2 \leq R\}$ for some $R > 0$. The boundary of a domain if denoted ∂D and the closure of D is denoted $\overline{D}$.

Definition 1.3 (Continuous function). For a subset $D \subset \mathbf{R}^d$, $C(D)$ denotes the set of real-valued *continuous functions* on D. Functions in $C(D)$ are not necessarily bounded. However, when D is bounded, the function in $C(\overline{D})$ are bounded.

Definition 1.4 (Norm). A *norm* $\|\cdot\|$ is a function from a real (or complex) vector space V to $\mathbf{R}^+$ such that: (i) $\|u\| = 0$ if and only if $u = 0$. (ii) $\|\lambda u\| = |\lambda|\|u\|$ for all $u \in V$ and $\lambda \in \mathbf{R}$ or $(\mathbf{C})$. and (iii) $\|u + v\| \leq \|u\| + \|v\|$ for all $u, v \in V$.

Definition 1.5 (Normed Vector Space). A *normed vector space* $(V, \|\cdot\|)$ is a vector space V with a norm $\|\cdot\|$.

Example 1.6. $(\mathbf{R}^n, \|\cdot\|_2)$ is a normed vector space.

Example 1.7. If D is a domain, then $(C(\overline{D}), \|\cdot\|_\infty)$ is a normed vector space with norm

$$\|u\|_\infty := \sup_{\mathbf{x} \in D} |u(\mathbf{x})|, \quad u \in C(\overline{D}). \tag{2}$$

Definition 1.8 (Complete). Let $(V, \|\cdot\|)$ be a normed vector space. A sequence $u_n \in V$, $n \in \mathbf{N}$ is called a *Cauchy sequence* if, for all $\varepsilon > 0$, there exist an $N \in \mathbf{N}$ such that

$$\|u_n - u_m\| < \varepsilon, \quad \forall n, m > N. \tag{3}$$

We say $(V, \|\cdot\|)$ is *complete* if, for every Cauchy sequence $u_n \in V$, there exist $u \in V$ such that $\|u_n - u\| \to 0$ as $n \to \infty$.

Note: Complete normed vector space is the normed vector space such that every cauchy sequence *converges* with respect to the norm.

Definition 1.9 (Banach space). A *Banach space* is a complete normed vector space.

Example 1.10. $(\mathbf{R}^n, \|\cdot\|_2)$ is a Banach space.

Example 1.11. If D is a bounded domain, then $(C(\overline{D}), \|\cdot\|_\infty)$ is a Banach space.

Definition 1.12 (Inner product). An inner product on a real vector space V is a function $\langle \cdot, \cdot \rangle : V \times V \to \mathbf{R}$ such that (i) $\langle u, u \rangle \geq 0$, $\forall u \in V$, and $\langle u, u \rangle = 0$ if and only if $u = 0$. (ii) $\langle u, v \rangle = \langle v, u \rangle$, $\forall u, v \in V$. and (iii) $\langle \lambda u + \mu v, w \rangle = \lambda \langle u, w \rangle + \mu \langle v, w \rangle$, $\forall u, v, w \in V$ and $\lambda, \mu \in \mathbf{R}$.

Definition 1.13 (Hilbert Space). A real valued Hilbert space H with inner product $\langle \cdot, \cdot \rangle_H$ is a Hilbert space if it is complete with respect to the induced norm $\|u\|_H := \langle u, u \rangle_H^{1/2}$.

Lemma 1.1 (Cauchy-Schwarz inequality). Let H be a Hilbert space with respect to the inner product $\langle \cdot, \cdot \rangle_H$. Then $|\langle u, v \rangle_H| \leq \|u\|_H \|v\|_H$ for all $u, v \in H$.

Definition 1.14 (Measurable Space, Measure Space). Let Ψ be a set with an associated sigma-algebra $\mathcal{G}$. Then we say $(\Psi, \mathcal{G})$ is a measurable space, and the sets in $\mathcal{G}$ are known as measurable sets. Given a measure π On a measurable space $(\Psi, \mathcal{G})$ (which assigns a measure to each set in $\mathcal{G}$), we call $(\Psi, \mathcal{G}, \pi)$ a measure space.

Note: The probability space $(\Omega, \mathcal{F}, \mathbf{P})$ is an example of measure space. The probability measure $\mathbf{P}$ has the special property that $\mathbf{P}(\Omega) = \int_\Omega 1 \, \mathrm{d}\mathbf{P}(w) = 1$.

Definition 1.15 (Borel sigma-algebra). When Ψ is a subset of $\mathbf{R}$ or $\mathbf{R}^n$ or is a Banach space, we can create a special sigma-algebra $\mathcal{B}(\Psi)$ called the *Borel sigma-algebra*. This is the smallest sigma-algebra containing all open subsets of Ψ.

Example 1.16. Let $D \subset \mathbf{R}^d$ be a domain. Then $(D, \mathcal{B}(D))$ is a measurable space. Standard Lebesgue measure is well defined on subsets of $(D, \mathcal{B}(D), \mathrm{Leb})$ is a measure space.

Definition 1.17 (L^p spaces on a domain). Let D be a domain, for $1 \leq p < \infty$, $L^p(D)$ is the se of (Borel-measurable) functions $u : D \to \mathbf{R}$ satisfying

$$\|u\|_{L^p(D)} := \left(\int_D |u(\mathbf{x})|^p \mathrm{d}\mathbf{x} \right)^{1/p} < \infty \tag{4}$$

Example 1.18 (Case $p = 2$). For any domain D, $L^2(D)$ is the set of square integrable functions $u : D \to \mathbf{R}$ satisfying

$$\|u\|_{L^2(D)} := \left(\int_D |u(\mathbf{x})|^2 \mathrm{d}\mathbf{x} \right)^{1/2} < \infty. \tag{5}$$

$L^2(D)$ is a Hilbert space with respect to the inner product

$$\langle u, v \rangle_{L^2(D)} := \int_D u(\mathbf{x}) v(\mathbf{x}) \mathrm{d}\mathbf{x}, \tag{6}$$

and by the Cauchy-Schwarz inequality we have

$$\langle u, v \rangle_{L^2(D)} \leq \|u\|_{L^2(D)} \|v\|_{L^2(D)}, \quad u, v \in L^2(D). \tag{7}$$